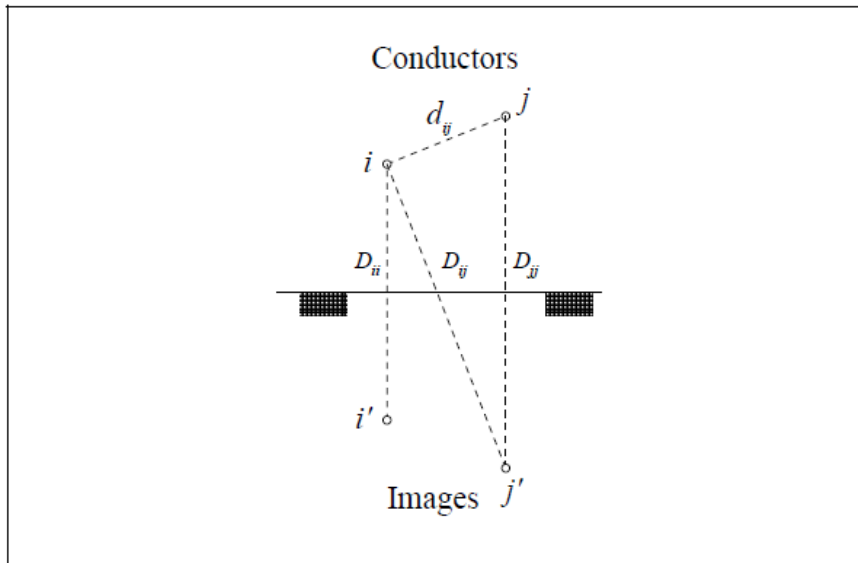


Exercise 1: Capacitance matrix

Two conductors of 100 mm diameter, conductor 1 is 5m above ground, conductor 2 is 1 m higher and directly above conductor 1.

Determine the capacitance matrix.

Note: The vacuum permittivity is $\epsilon_0 = 8.8542 \times 10^{-12} \text{ Fm}^{-1}$



$$\mathbf{A} = \text{matrix with elements } A_{ij} = \ln \frac{D_{ij}}{d_{ij}}$$

$$d_{ii} = R_i = \text{radius of conductor } i$$

$$D_{ii} = 2h_i = \text{double height of conductor } i$$

$$A_{11} = \ln \frac{10}{0.05} = 5.298 \quad A_{12} = \ln \frac{11}{1} = 2.398$$

$$A_{21} = A_{12} \quad A_{22} = \ln \frac{12}{0.05} = 5.481$$

$$\mathbf{A} = \begin{bmatrix} 5.298 & 2.398 \\ 2.398 & 5.481 \end{bmatrix} \quad \mathbf{A}^{-1} = \frac{1}{23.288} \begin{bmatrix} 5.481 & -2.398 \\ -2.398 & 5.298 \end{bmatrix}$$

$$\mathbf{C} = 2\pi\epsilon_0 \mathbf{A}^{-1} \quad \text{Fm}^{-1}$$

$$\mathbf{C} = 2\pi \times 8.8542 \times 10^{-12} \mathbf{A}^{-1} \text{ Fm}^{-1}$$

$$= \begin{bmatrix} 13.09 & -5.73 \\ -5.73 & 12.66 \end{bmatrix} \text{ pFm}^{-1}$$

(McLean 2012)

Exercise 2: Railway transmission line and pasture fence

There is a pasture fence (f), that is capacitively influenced by a two-phase (a,b) 110 kV railway transmission line of $f = 16,7$ Hz.

The capacitance matrix of the system is given by

$$c'_{abf} = \begin{bmatrix} 8,910 & -1,807 & -0,048 \\ -1,807 & 8,135 & -0,091 \\ -0,048 & -0,091 & 11,508 \end{bmatrix} \frac{nF}{km}$$

- Calculate the touch voltage of the fence
- Calculate the touch current of the fence
- What is the maximum length of the fence in parallel to the line, so that a touch current of 2 mA is not exceeded?

(Moser 2010)

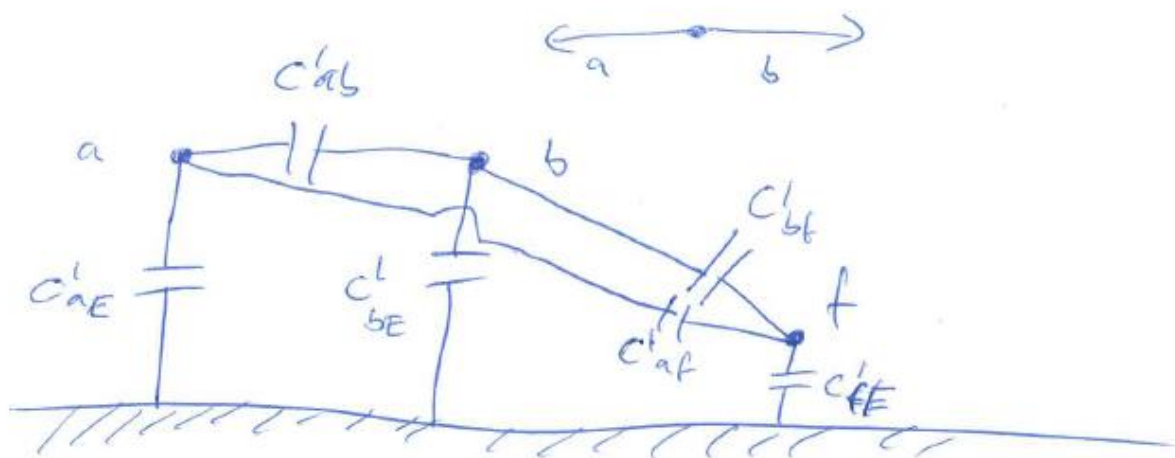
$$U_n = 110 \text{ kV}$$

$$f = 16,7 \text{ Hz}$$

Railway line:

$$U_{V1} = 55 \text{ kV}$$

$$U_{V2} = -55 \text{ kV}$$



$$C'_{abf} = \begin{pmatrix} C'_{aa} & C'_{ab} & C'_{af} \\ C'_{ba} & C'_{bb} & C'_{bf} \\ C'_{fa} & C'_{fb} & C'_{ff} \end{pmatrix}$$

Kapazitäts-
koeffizienten-
matrix

$$a) \begin{pmatrix} \Delta I_a' \\ \Delta I_b' \\ \Delta I_f' \end{pmatrix} = j\omega \cdot C_{abf}' \cdot \begin{pmatrix} \underline{U}_a \\ \underline{U}_b \\ \underline{U}_c \end{pmatrix}$$

Insulated wire: There is no current flow

$$\Rightarrow \Delta I_f' = 0$$

$$\begin{aligned} \Delta I_f' = 0 &= j\omega \left(-0,048 \underline{U}_a - 0,091 \underline{U}_b + 11,508 \underline{U}_f \right) \frac{\mu F}{\text{km}} \\ &= 0,043 \underline{U}_a + 11,508 \underline{U}_f \end{aligned}$$

$$\Leftrightarrow -0,043 \underline{U}_a = 11,508 \underline{U}_f$$

$$U_f = \left| - \frac{0,043 \underline{U}_a}{11,508} \right| = 205 \text{ V}$$

b) Berührung über vernachlässigbaren Körperwiderstand $\Rightarrow \underline{U}_f = 0$

(The body resistance is neglected $\Rightarrow U_f = 0$)

$$\begin{aligned}\Delta I_f' &= j\omega \left(-0,048 \underline{U}_a - 0,091 \underline{U}_b \right) \frac{nF}{km} \\ &= j\omega 0,043 \underline{U}_a \frac{nF}{km} \\ &= j2\pi \cdot 16,7 \text{ Hz} \cdot 55 \text{ kV} \cdot \frac{nF}{km} \\ &= j 0,25 \frac{\text{mA}}{\text{km}}\end{aligned}$$

$$\Rightarrow I_f' = 0,25 \frac{\text{mA}}{\text{km}}$$

c) $I_{f,\max} = I_f' \cdot l_{\max}$

$$l_{\max} = \frac{I_{f,\max}}{I_f'} = \frac{2 \text{ mA}}{0,25 \frac{\text{mA}}{\text{km}}} = 8 \text{ km}$$